

CUSTOMER INTELLIGENCE DASHBOARD USING MACHINE LEARNING, FASTAPI AND GENERATIVE AI

Comprehensive Project Report

Abstract

This project develops an end-to-end Customer Intelligence Dashboard that combines Machine Learning, FastAPI, Power BI, and Google's Gemini Generative AI. The system performs customer segmentation, spending prediction, subscription prediction, and AI-powered business insight generation. The objective is to transform raw customer shopping data into actionable intelligence that supports marketing, customer retention, and revenue growth decisions.

Problem Statement

Modern retail businesses generate large volumes of customer transaction data but often struggle to transform this data into actionable insights. This project addresses that challenge through an intelligent analytics platform capable of identifying customer segments, predicting spending behavior, estimating subscription likelihood, and generating AI-driven recommendations.

Objectives

- Customer Segmentation using K-Means Clustering
- Spending Prediction using Regression Models
- Subscription Prediction using Classification Models
- REST API Development using FastAPI
- Interactive Dashboard Development
- AI Insight Generation using Gemini API
- Business Intelligence Visualization using Power BI

Technology Stack

Python, Pandas, NumPy, Scikit-Learn, Imbalanced-Learn (SMOTE), FastAPI, Uvicorn, HTML, CSS, JavaScript, Power BI, Google Gemini API, Joblib.

Dataset Description

The project uses a retail customer behavior dataset containing 3900 records. Features include age, purchase amount, review rating, previous purchases, category preferences, subscription status, shipping preferences, and seasonal behavior patterns.

Data Preprocessing

Data cleaning, validation, feature selection, and feature engineering were performed. New behavioral features such as Purchase Frequency Days and Purchases Per Year were created to improve model performance and customer understanding.

Feature Engineering

Purchase Frequency Days estimates how often customers make purchases. Purchases Per Year standardizes buying activity and helps identify loyal and highly active customers. These features were used throughout segmentation and prediction models.

Customer Segmentation

K-Means clustering was applied after feature scaling. Elbow Method and Silhouette Analysis were used to determine the optimal number of clusters. Six meaningful customer segments were identified:

1. Premium Loyal Customers
2. Low-Spend Senior Customers
3. Young Satisfied Shoppers
4. Loyal Repeat Customers
5. High-Spend Occasional Buyers
6. Frequent Active Buyers

Segmentation Results

Silhouette analysis indicated meaningful cluster separation. Segment profiles showed clear differences in age, spending behavior, review ratings, and purchase frequency, enabling targeted marketing strategies.

Spending Prediction Model

Multiple regression models were evaluated. Linear Regression achieved the best performance and was selected for deployment.

Results:

RMSE = 23.73

MAE = 20.70

$$R^2 = -0.0067$$

The model predicts expected customer spending using demographic and behavioral attributes.

Subscription Prediction Model

Subscription prediction was treated as a binary classification problem. Because the dataset was imbalanced (73% No, 27% Yes), SMOTE was applied.

Best Model: Logistic Regression

Accuracy = 59.49%

Precision = 40.04%

Recall = 100%

F1 Score = 57.18%

ROC-AUC = 72.68%

The model estimates the likelihood of a customer subscribing to a service.

FastAPI Backend Development

The backend exposes machine learning models through REST APIs.

Implemented Endpoints:

GET /

GET /model-status

POST /segment-customer

POST /predict-spending

POST /predict-subscription

POST /generate-insight

FastAPI automatically generates interactive Swagger documentation for testing APIs.

Frontend Dashboard

A web dashboard was developed using HTML, CSS, and JavaScript. Users can enter customer information and obtain segmentation results, spending forecasts, subscription predictions, and AI-generated business insights in real time.

Generative AI Integration

Google Gemini API was integrated into the solution. Instead of returning only numerical predictions, the AI layer converts predictions into business-oriented insights including customer analysis, recommendations, and marketing strategies.

AI Insight Workflow

Input:

- Segment Name
- Predicted Spending
- Subscription Prediction
- Subscription Probability

Output:

- Customer Insight
- Business Recommendation
- Marketing Strategy

This makes the platform useful for both technical and non-technical stakeholders.

Power BI Dashboard

The Power BI dashboard includes KPI cards, subscription distribution, revenue by category, sales by category, revenue by age group, customer counts, and customer trend analysis. Interactive filters enable business users to explore customer behavior dynamically.

Results and Findings

The project successfully integrated machine learning, business intelligence, APIs, and generative AI into a single platform.

Major Findings:

- Premium Loyal Customers demonstrate high retention potential.
- Customer segments differ significantly in spending behavior.
- Subscription probability provides a useful marketing targeting metric.
- AI-generated recommendations improve interpretability of predictions.

Business Impact

The system enables personalized marketing campaigns, customer retention programs, targeted promotions, subscription optimization, and data-driven decision-making. It helps businesses reduce manual analysis and improve operational efficiency.

Challenges Faced

- Class imbalance in subscription prediction
- Feature engineering for customer behavior
- Integration of machine learning with APIs
- Gemini API configuration and deployment
- Frontend-backend communication

Future Scope

- Churn Prediction
- Product Recommendation Systems
- Cloud Deployment (AWS/Azure/GCP)
- Real-Time Streaming Analytics
- Retrieval-Augmented Generation (RAG)
- Multi-Channel Customer Intelligence
- Advanced Deep Learning Models

Conclusion

The Customer Intelligence Dashboard demonstrates how Machine Learning and Generative AI can work together to transform customer data into actionable business intelligence. The project successfully delivers customer segmentation, spending prediction, subscription prediction, and AI-powered recommendations through a modern web-based analytics platform.

References

1. Scikit-Learn Documentation
2. FastAPI Documentation
3. Google Gemini API Documentation
4. Power BI Documentation
5. Pandas Documentation
6. NumPy Documentation
7. Imbalanced-Learn Documentation

Screenshots to Insert Before Final Submission

- Project Folder Structure
- Feature Engineering Output
- Elbow Method Graph
- Silhouette Analysis Results
- Customer Segmentation Output
- Spending Prediction Model Results
- Subscription Prediction Model Results
- Confusion Matrix
- Swagger API Documentation
- Segment Customer API Output
- Predict Spending API Output
- Predict Subscription API Output
- Generate Insight API Output
- Customer Dashboard Screens
- Power BI Dashboard